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Eden et al.

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(54) **REFRIGERATOR APPLIANCE WITH SMART DRAWERS**

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See application file for complete search history.

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F25D 25/02	(2006.01)
G06V 20/68	(2022.01)
G08B 21/18	(2006.01)

(57) **ABSTRACT**

Methods of operating a refrigerator appliance are provided. The refrigerator appliance includes a cabinet defining a food storage chamber with a drawer slidably mounted within the food storage chamber. The refrigerator appliance also includes a sensor operable to detect an atmospheric condition within the food storage chamber of the drawer and a camera assembly positioned and configured for monitoring the drawer. The methods generally include obtaining an image using the camera assembly and analyzing the image to identify a first food item and a second food item in the food storage chamber of the drawer.

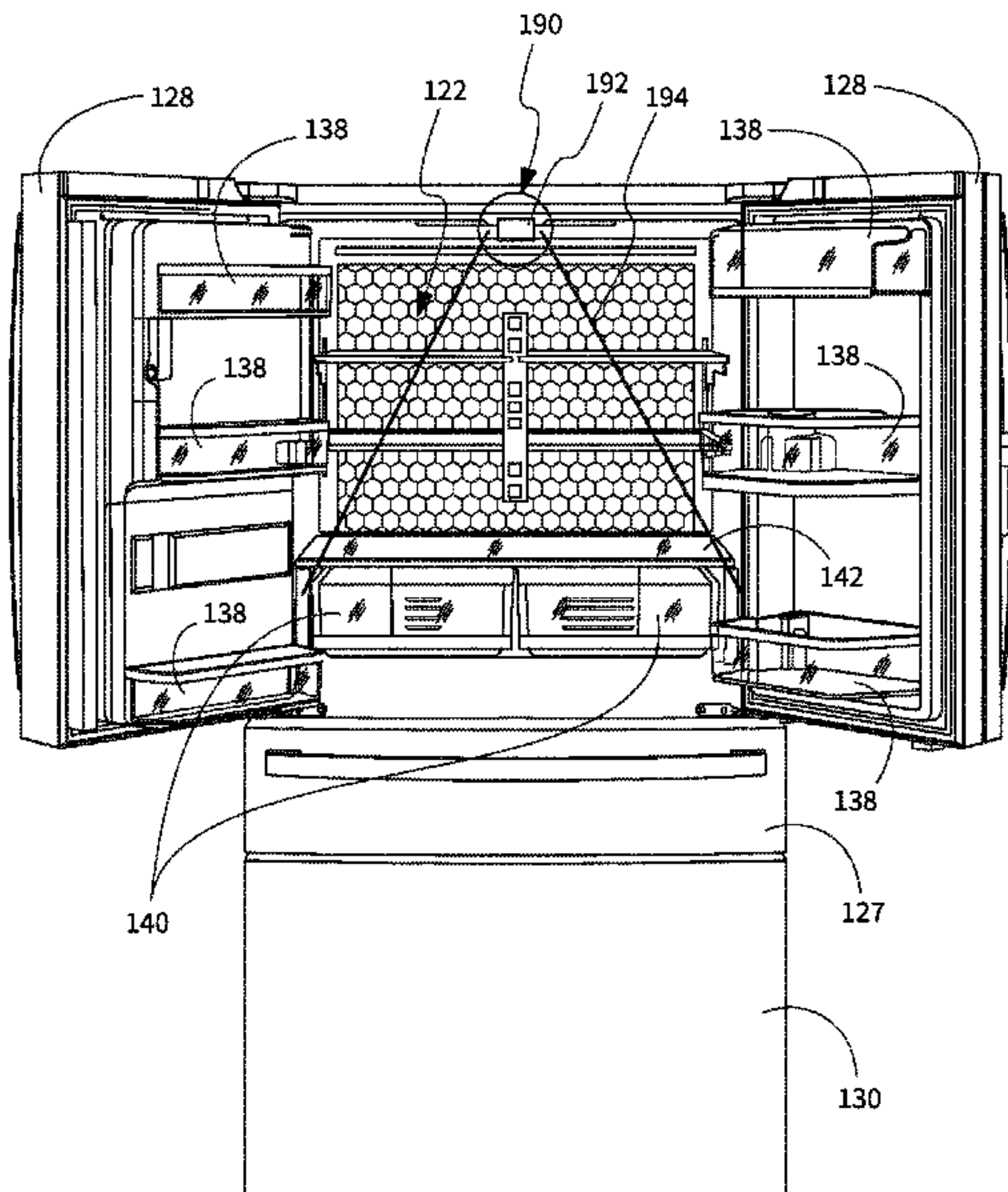
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(2013.01); **F25D 25/025** (2013.01); **G06V**
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14 Claims, 11 Drawing Sheets



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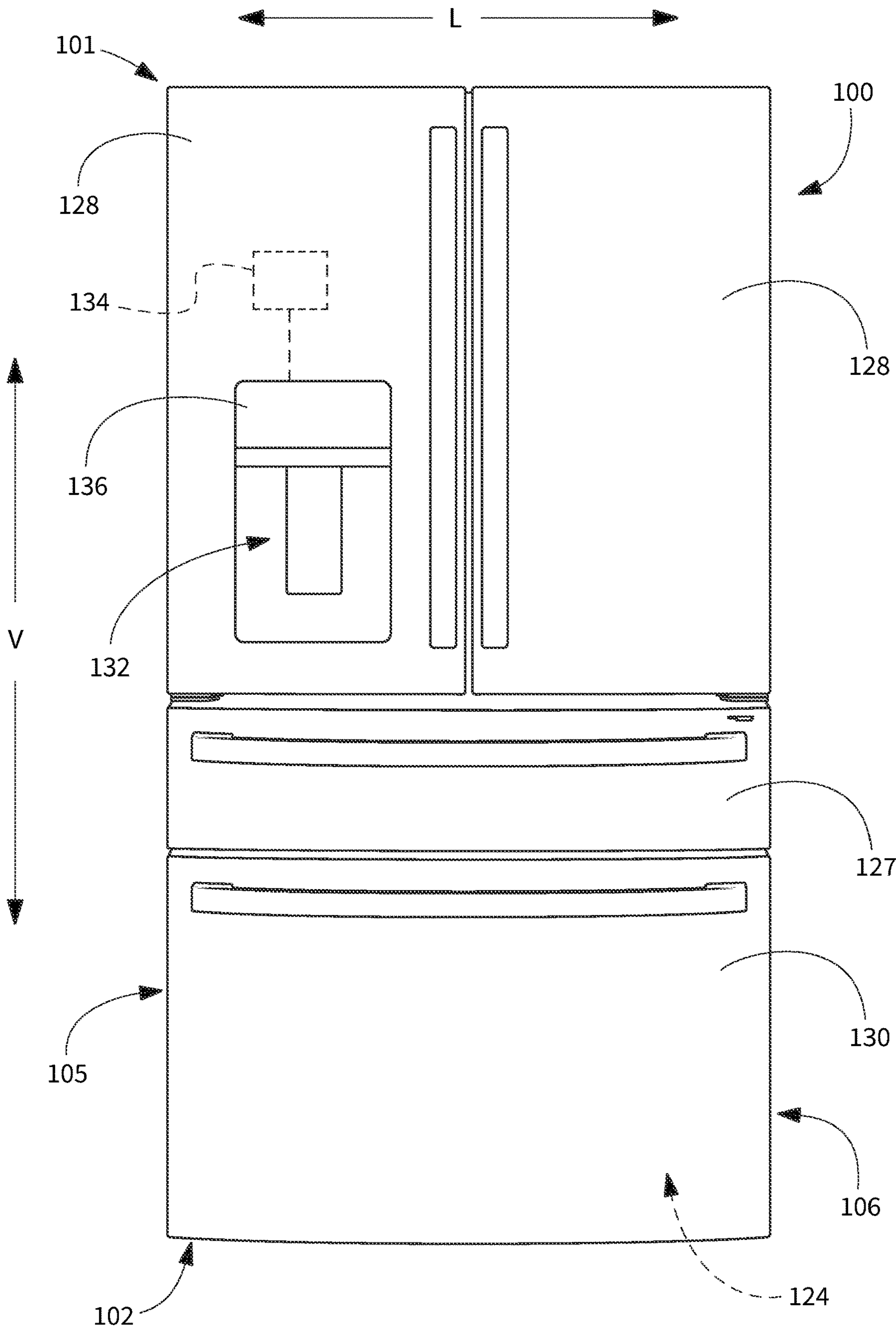


FIG. 1

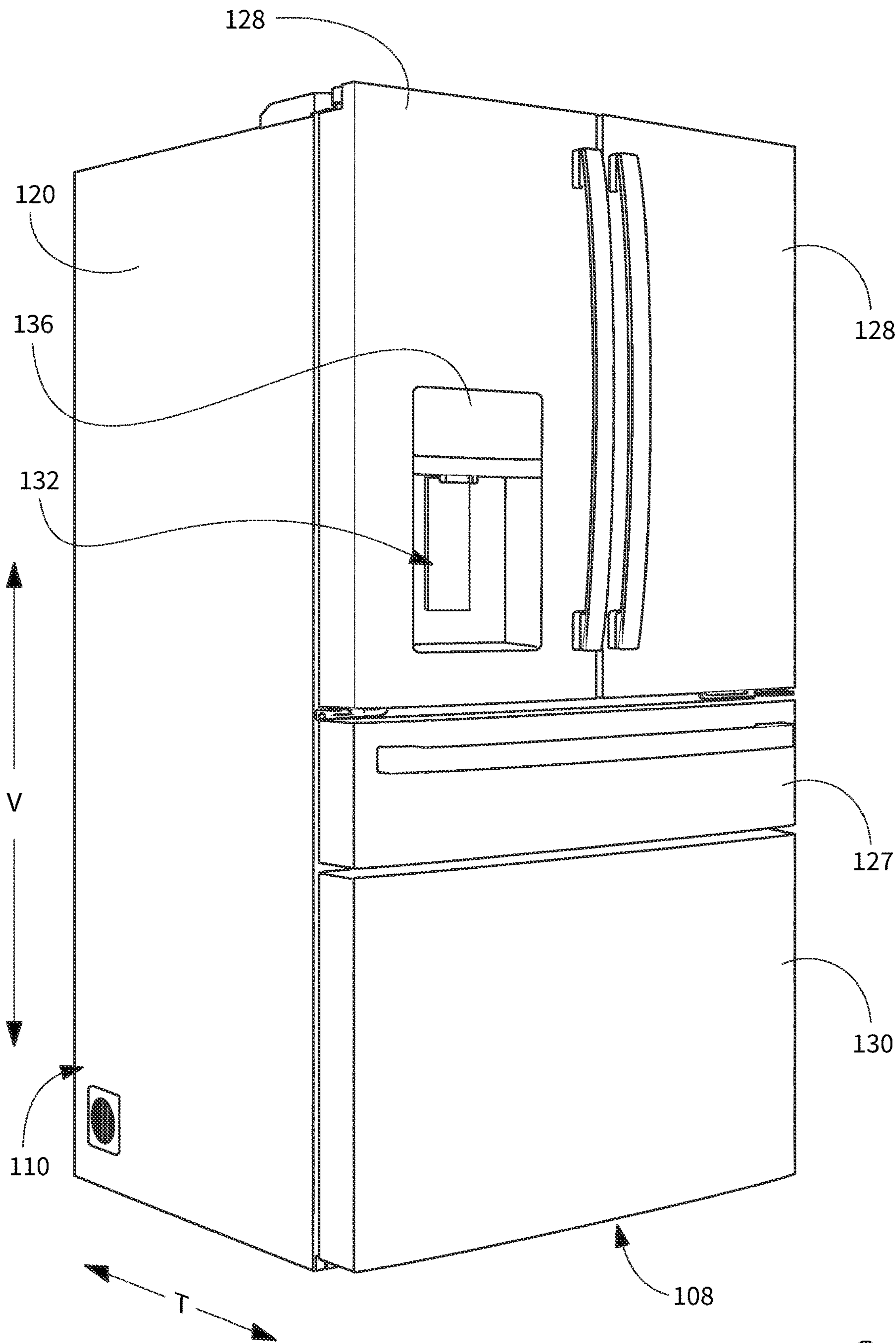


FIG. 2

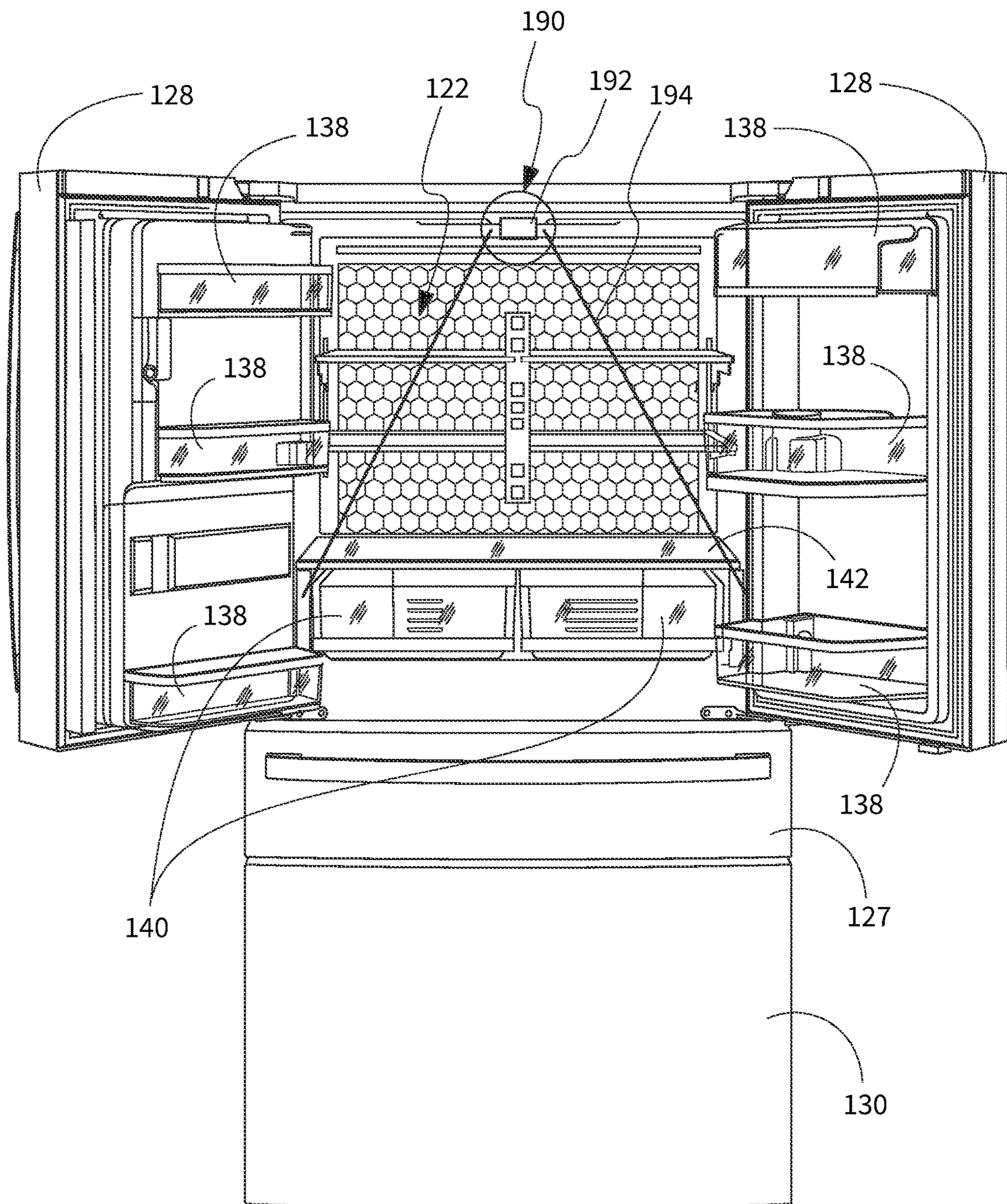


FIG. 3

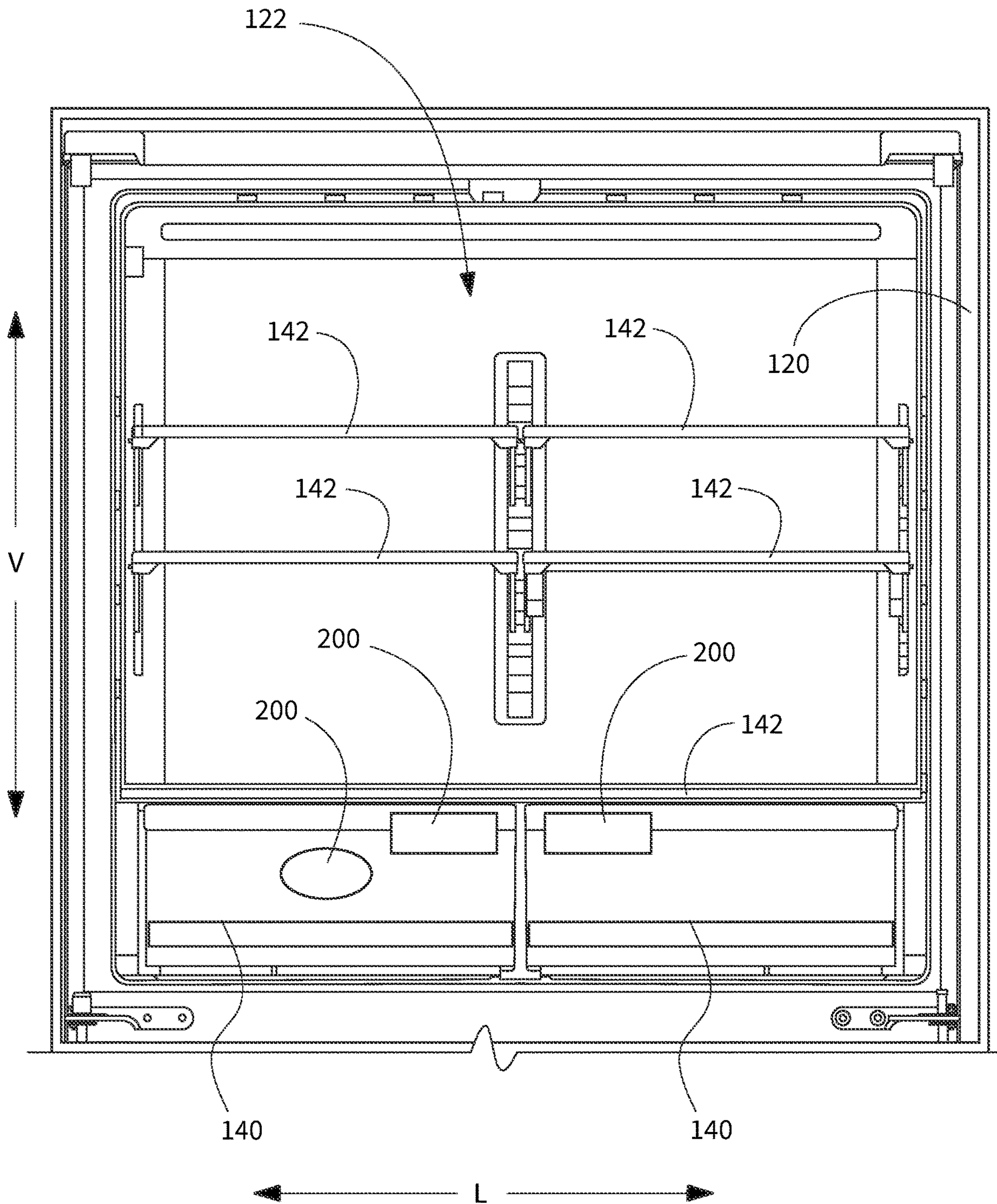


FIG. 4

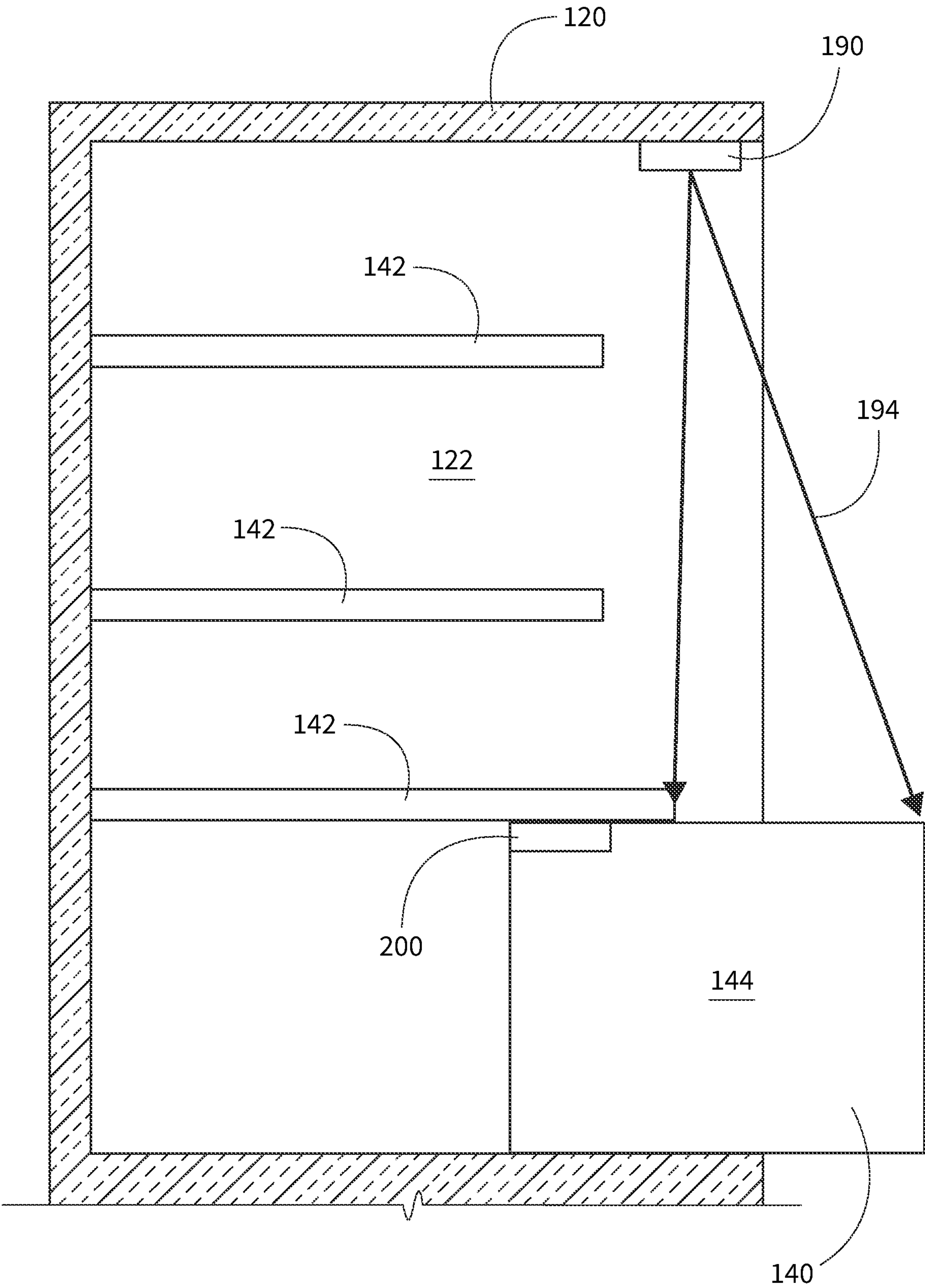


FIG. 5

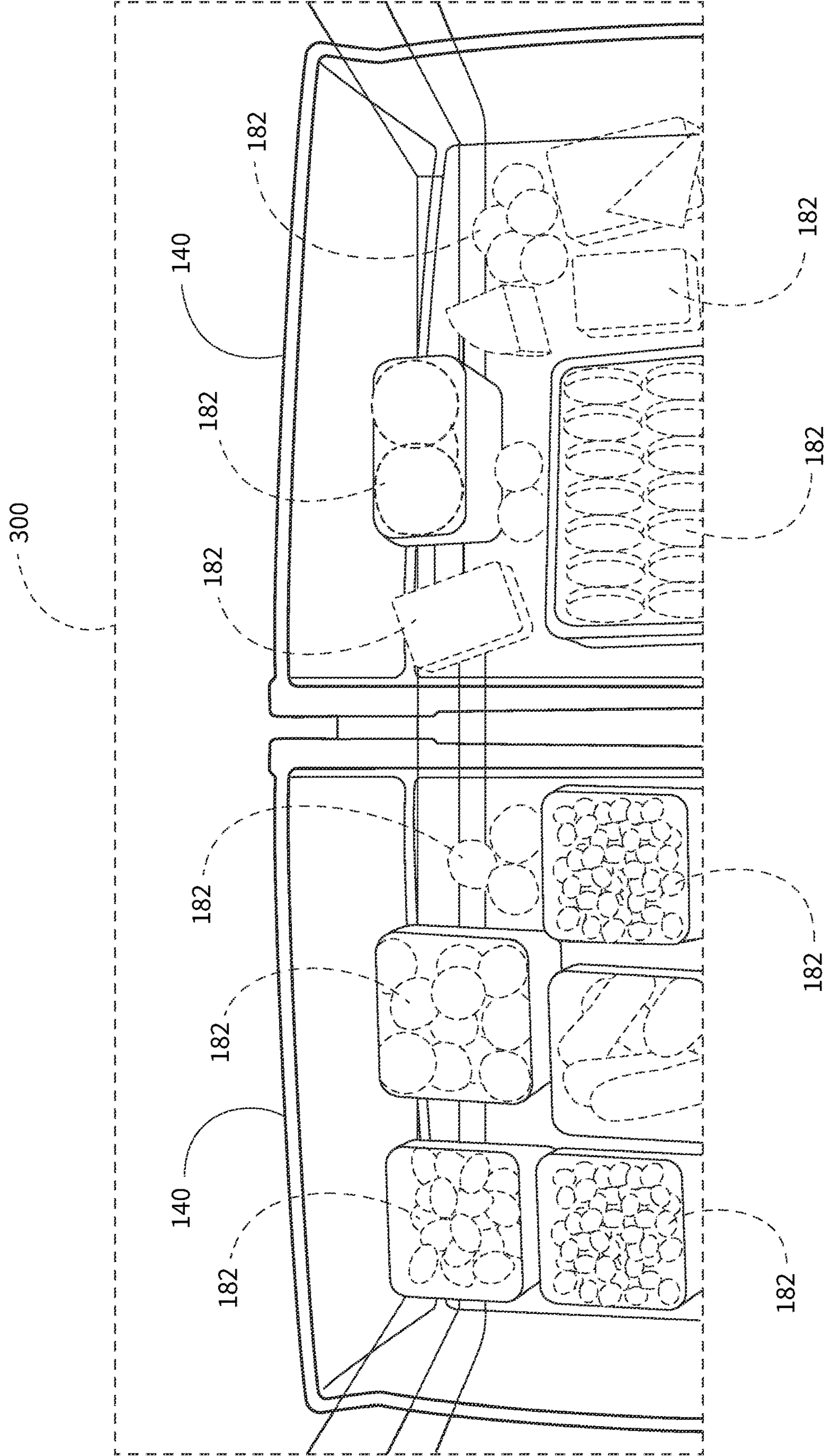


FIG. 6

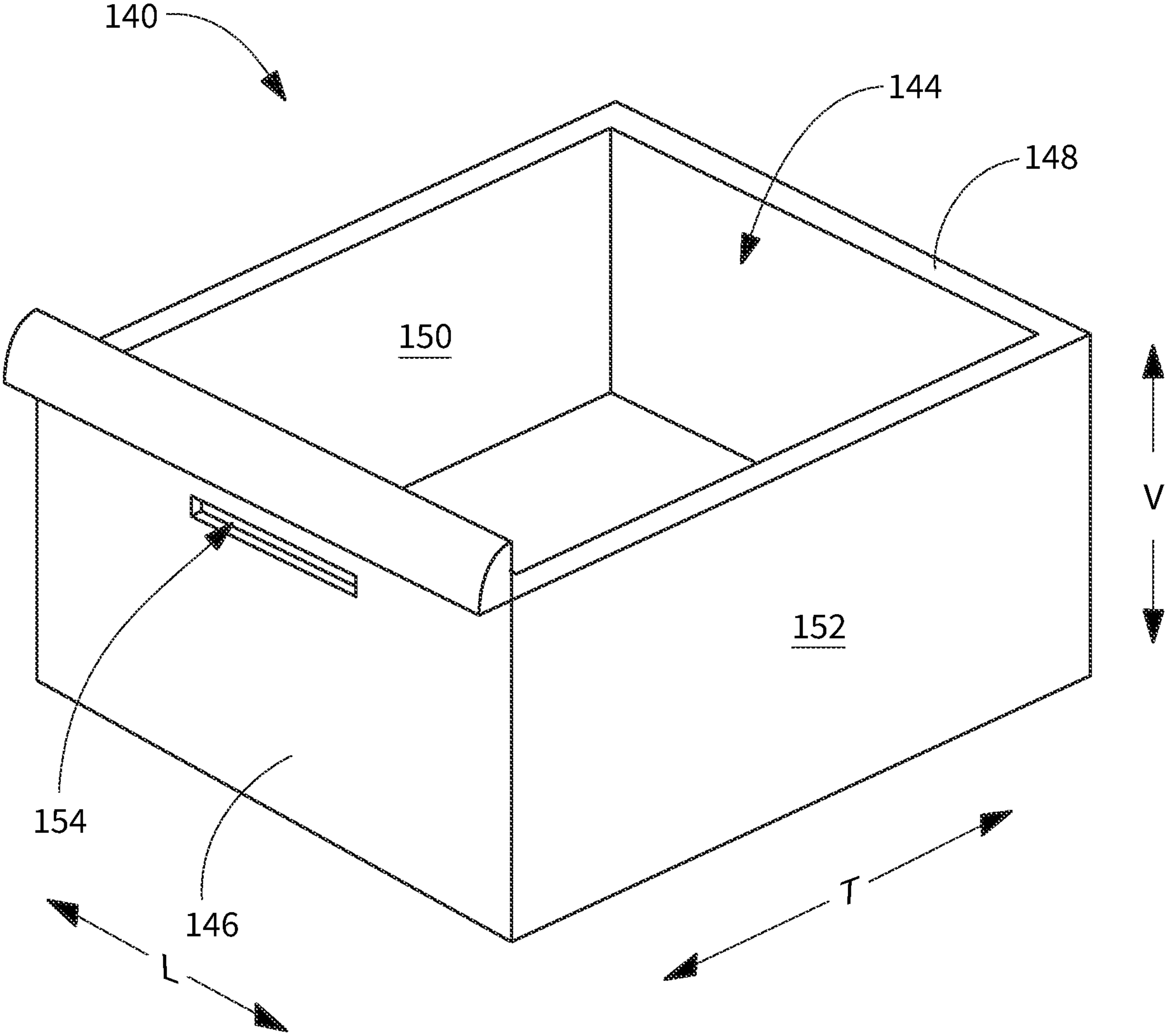


FIG. 7

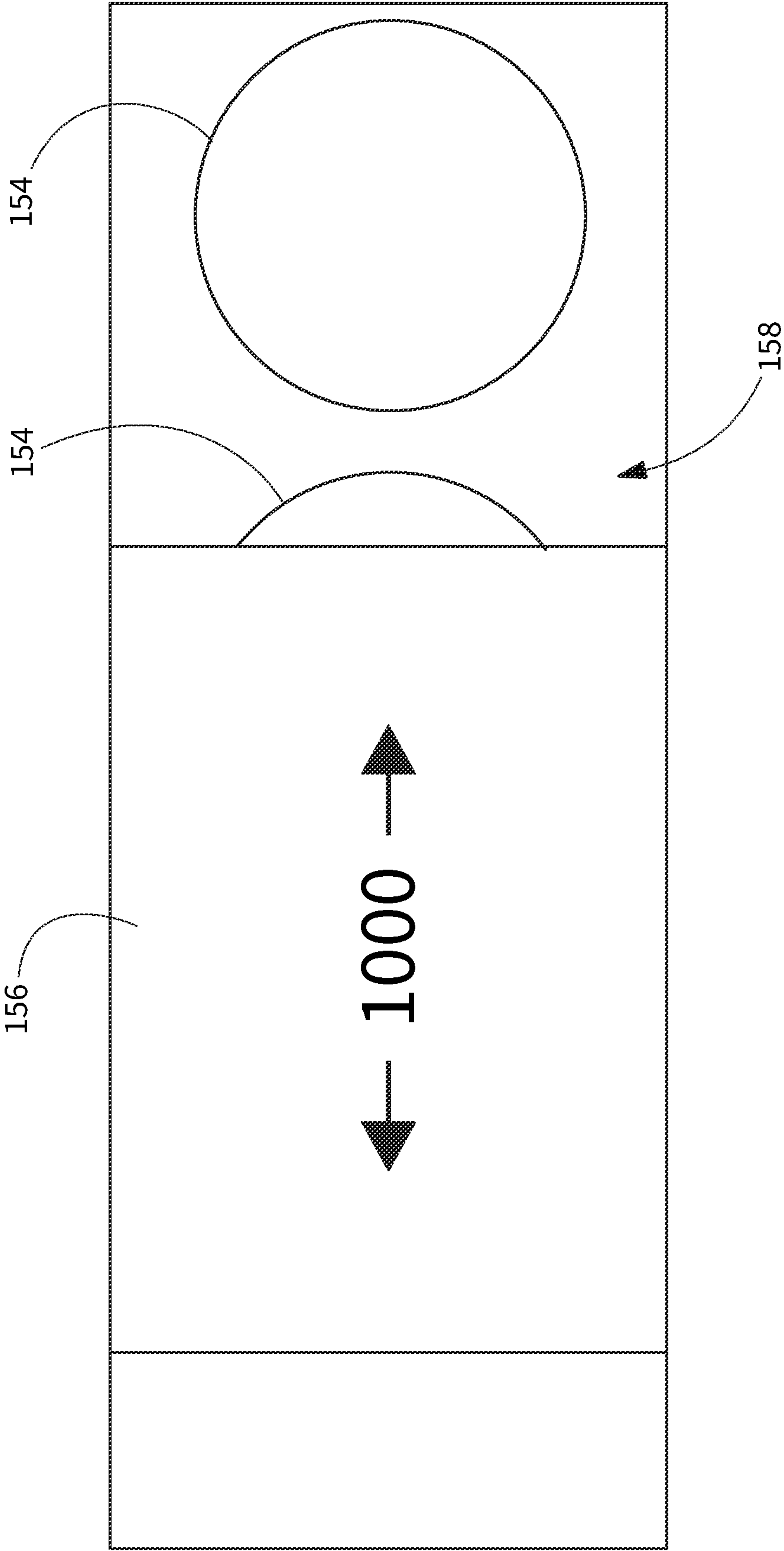


FIG. 8

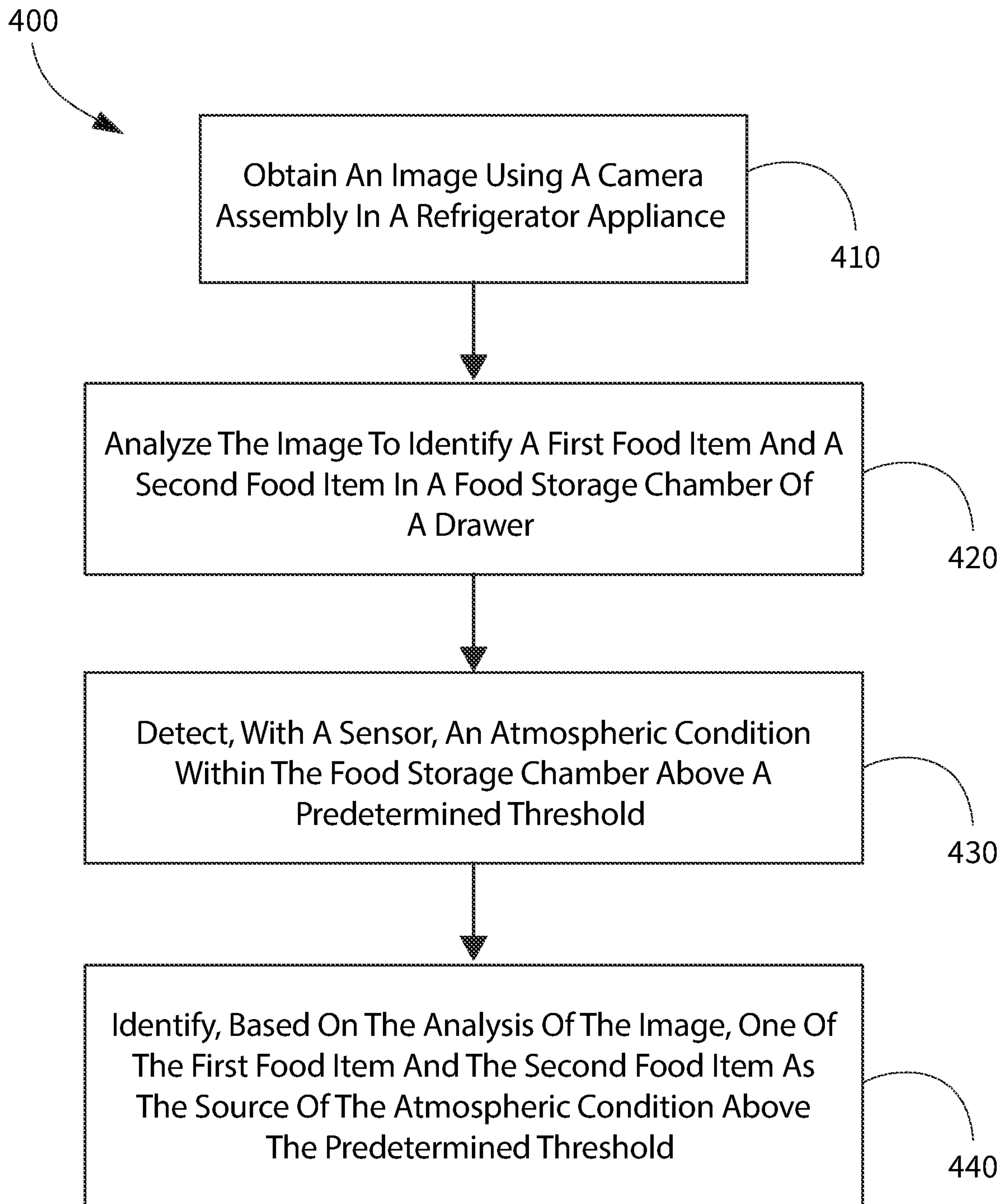


FIG. 9

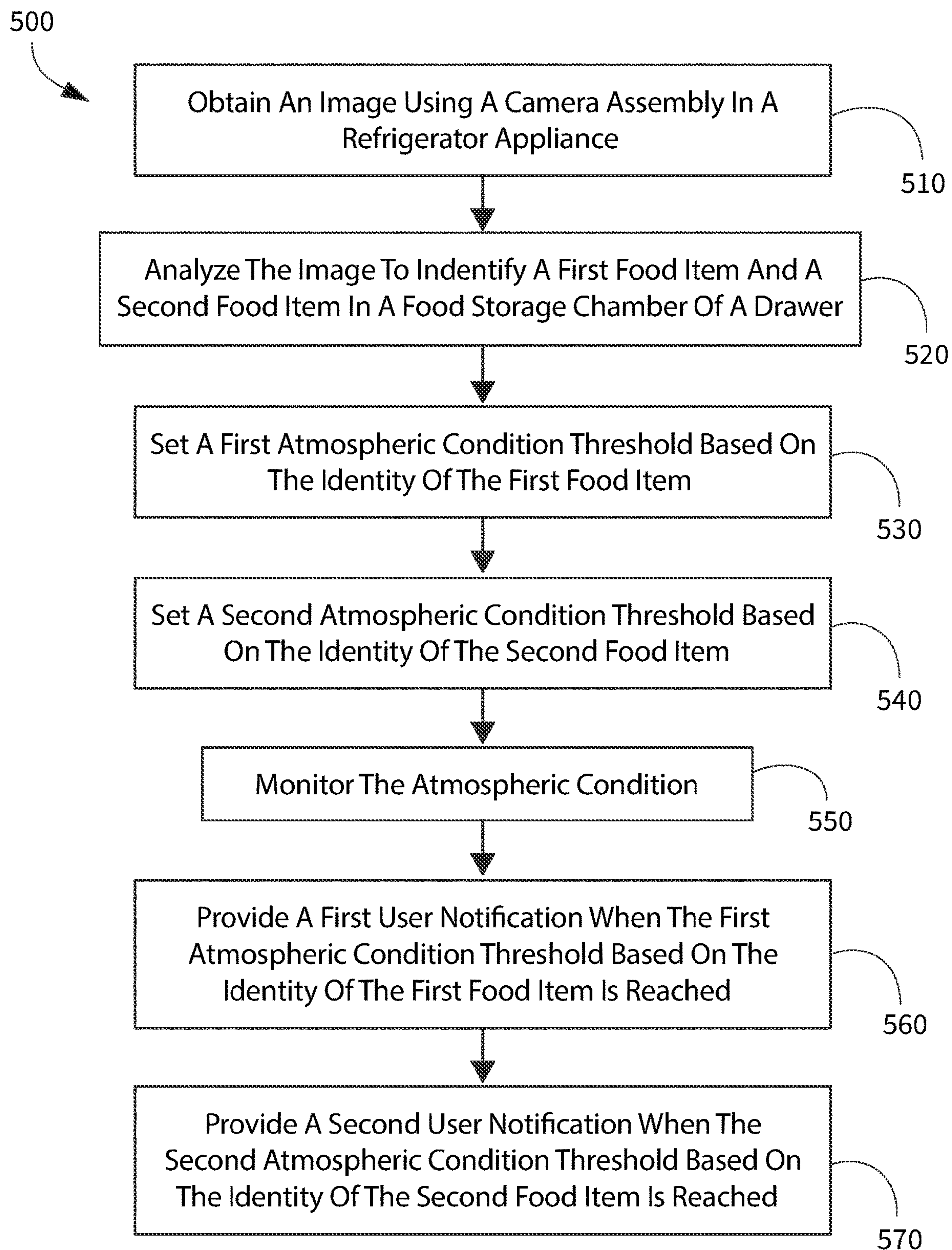


FIG. 10

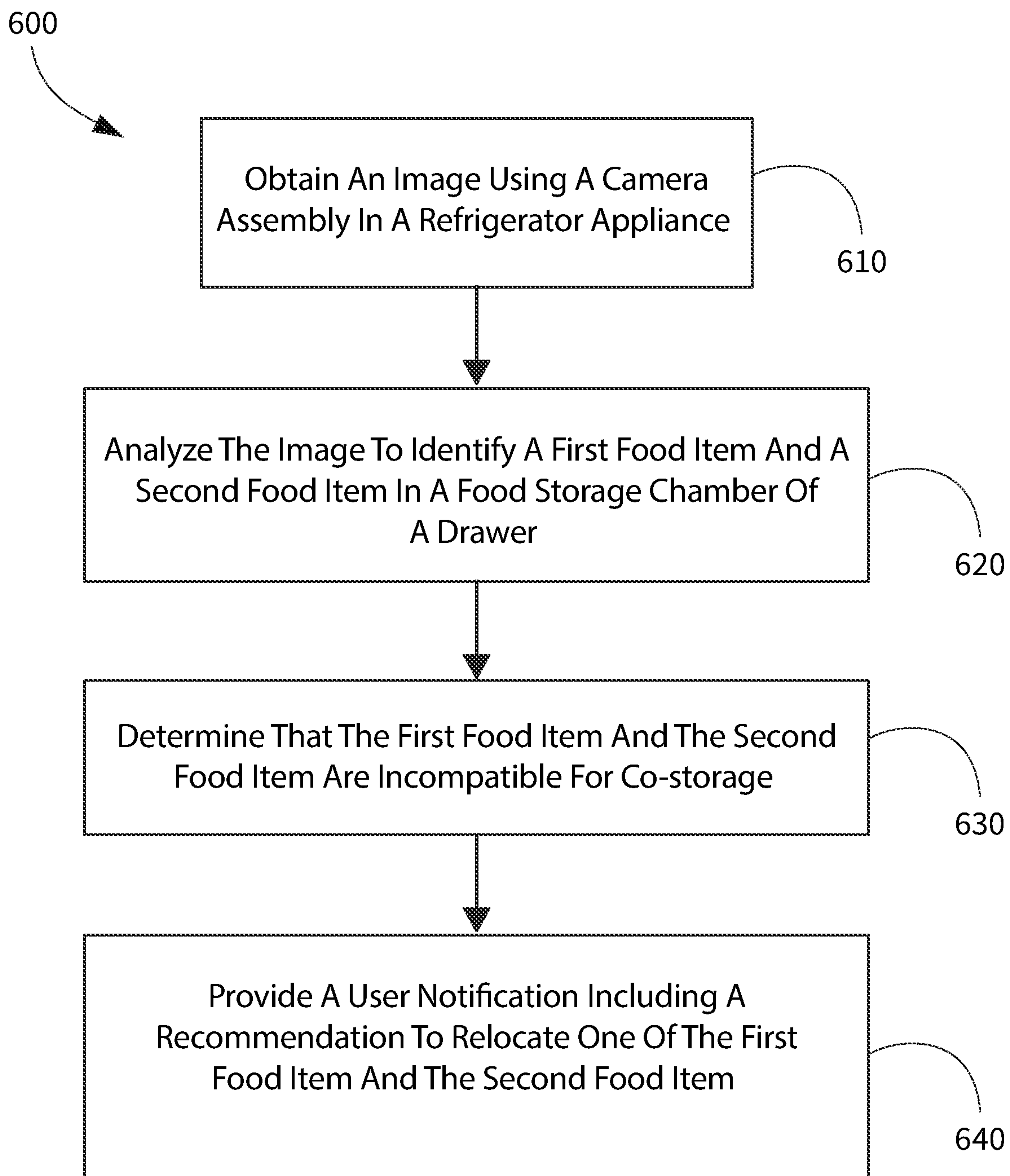


FIG. 11

REFRIGERATOR APPLIANCE WITH SMART DRAWERS

FIELD OF THE INVENTION

The present subject matter relates generally to refrigerator appliances, and more particularly to systems and methods for managing the status of items, such as produce items, stored in such refrigerator appliances.

BACKGROUND OF THE INVENTION

Refrigerator appliances generally include a cabinet that defines a chilled chamber. A wide variety of food items may be stored within the chilled chamber. The low temperature of the chilled chamber relative to ambient atmosphere assists with increasing a shelf life of the food items stored within the chilled chamber.

Produce items, e.g., fruits and vegetables, stored in a refrigerator appliance go through various physical and chemical changes over time, e.g., ripening. Various different produce items may be incompatible with each other, such as may have differing storage requirements. For example, the optimal temperature, humidity and/or atmospheric composition for one produce item may differ from that of another produce item. Different produce items may also be incompatible for storage together due to differences in the changes that occur in each over time while stored, such as one produce item may generate a certain atmospheric chemical while ripening that is deleterious for other produce items.

Accordingly, a refrigerator appliance with systems for improved inventory management would be useful. More particularly, a refrigerator appliance that includes a produce inventory management system that is capable of monitoring produce inventory and tracking the status of such items during storage would be useful.

BRIEF DESCRIPTION OF THE INVENTION

Aspects and advantages of the invention will be set forth in part in the following description, or may be apparent from the description, or may be learned through practice of the invention.

In an exemplary embodiment, a method of operating a refrigerator appliance is provided. The refrigerator appliance includes a cabinet defining a food storage chamber with a drawer slidably mounted within the food storage chamber. The drawer is slidable between a closed position and an open position. The drawer includes a plurality of walls defining a food storage compartment. The refrigerator appliance also includes a sensor operable to detect an atmospheric condition within the food storage chamber of the drawer and a camera assembly positioned and configured for monitoring the drawer. The method includes obtaining an image using the camera assembly and analyzing the image to identify a first food item and a second food item in the food storage chamber of the drawer. The method also includes detecting, with the sensor, an atmospheric condition within the food storage chamber above a predetermined threshold, and identifying, based on the analysis of the image, one of the first food item and the second food item as the source of the atmospheric condition above the predetermined threshold.

In another exemplary embodiment, a method of operating a refrigerator appliance is provided. The refrigerator appliance includes a cabinet defining a food storage chamber with a drawer slidably mounted within the food storage chamber. The drawer is slidable between a closed position and an open

position. The drawer includes a plurality of walls defining a food storage compartment. The refrigerator appliance also includes a sensor operable to detect an atmospheric condition within the food storage chamber of the drawer and a camera assembly positioned and configured for monitoring the drawer. The method includes obtaining an image using the camera assembly and analyzing the image to identify a first food item and a second food item in the food storage chamber of the drawer. The method also includes setting a first threshold for an atmospheric condition based on the identity of the first food item and setting a second threshold for the atmospheric condition based on the identity of the second food item. The method further includes monitoring the atmospheric condition by the sensor. The method also includes providing a first user notification when the atmospheric condition reaches the first threshold and providing a second user notification when the atmospheric condition reaches the second threshold.

In still another exemplary embodiment, a method of operating a refrigerator appliance is provided. The refrigerator appliance includes a cabinet defining a food storage chamber with a drawer slidably mounted within the food storage chamber. The drawer is slidable between a closed position and an open position. The drawer includes a plurality of walls defining a food storage compartment. The refrigerator appliance also includes a sensor operable to detect an atmospheric condition within the food storage chamber of the drawer and a camera assembly positioned and configured for monitoring the drawer. The method includes obtaining an image using the camera assembly and analyzing the image to identify a first food item and a second food item in the food storage chamber of the drawer. The method also includes determining that the first food item and the second food item are incompatible for co-storage and providing a user notification including a recommendation to relocate one of the first food item and the second food item.

These and other features, aspects and advantages of the present invention will become better understood with reference to the following description and appended claims. The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate embodiments of the invention and, together with the description, serve to explain the principles of the invention.

BRIEF DESCRIPTION OF THE DRAWINGS

A full and enabling disclosure of the present invention, including the best mode thereof, directed to one of ordinary skill in the art, is set forth in the specification, which makes reference to the appended figures.

FIG. 1 provides a front view of a refrigerator appliance according to an exemplary embodiment of the present subject matter.

FIG. 2 provides a perspective view of the refrigerator appliance of FIG. 1.

FIG. 3 provides a front view of the refrigerator appliance of FIG. 1 with doors thereof in an open position.

FIG. 4 provides a front view of a portion of a refrigerator appliance according to one or more exemplary embodiments of the present subject matter.

FIG. 5 provides a section view of the portion of the refrigerator appliance of FIG. 4.

FIG. 6 illustrates an exemplary image of storage drawers of a refrigerator appliance, such as the exemplary refrigerator appliance of FIG. 1, in an open position and contents of the drawers, which image may be captured by a camera assembly in the refrigerator appliance.

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FIG. 7 provides a perspective view of a drawer which may be incorporated into a refrigerator appliance in one or more exemplary embodiments of the present subject matter.

FIG. 8 provides an enlarged view of a portion of the drawer of FIG. 7.

FIG. 9 provides a flow diagram of an exemplary method for operating a refrigerator appliance according to one or more exemplary embodiments of the present subject matter.

FIG. 10 provides a flow diagram of another exemplary method for operating a refrigerator appliance according to one or more additional exemplary embodiments of the present subject matter.

FIG. 11 provides a flow diagram of still another exemplary method for operating a refrigerator appliance according to one or more additional exemplary embodiments of the present subject matter.

DETAILED DESCRIPTION

Reference now will be made in detail to embodiments of the invention, one or more examples of which are illustrated in the drawings. Each example is provided by way of explanation of the invention, not limitation of the invention. In fact, it will be apparent to those skilled in the art that various modifications and variations can be made in the present invention without departing from the scope or spirit of the invention. For instance, features illustrated or described as part of one embodiment can be used with another embodiment to yield a still further embodiment. Thus, it is intended that the present invention covers such modifications and variations as come within the scope of the appended claims and their equivalents.

FIG. 1 is a front view of an exemplary embodiment of a refrigerator appliance 100. FIG. 2 is a perspective view of the refrigerator appliance 100. FIG. 3 is a front view of the refrigerator appliance 100 with fresh food doors 128 thereof in an open position. Refrigerator appliance 100 extends between a top 101 and a bottom 102 along a vertical direction V. Refrigerator appliance 100 also extends between a first side 105 and a second side 106 along a lateral direction L. As shown in FIG. 2, a transverse direction T may additionally be defined perpendicular to the vertical and lateral directions V and L. Refrigerator appliance 100 extends along the transverse direction T between a front portion 108 and a back portion 110.

Refrigerator appliance 100 includes a cabinet or housing 120 defining an upper fresh food chamber 122 (FIG. 3) and a lower freezer chamber or frozen food storage chamber 124 arranged below the fresh food chamber 122 along the vertical direction V. In some embodiments, an auxiliary food storage chamber (not shown) may be positioned between the fresh food storage chamber 122 and the frozen food storage chamber 124, e.g., along the vertical direction V. Because the frozen food storage chamber 124 is positioned below the fresh food storage chamber 122, refrigerator appliance 100 is generally referred to as a bottom mount refrigerator. In the exemplary embodiment, housing 120 also defines a mechanical compartment (not shown) for receipt of a sealed cooling system (not shown). Using the teachings disclosed herein, one of skill in the art will understand that the present invention can be used with other types of refrigerators (e.g., side-by-sides) as well. Consequently, the description set forth herein is for illustrative purposes only and is not intended to limit the invention in any aspect.

Refrigerator doors 128 are each rotatably hinged to an edge of housing 120 for accessing fresh food chamber 122. It should be noted that while two doors 128 in a “French

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door” configuration are illustrated, any suitable arrangement of doors utilizing one, two or more doors is within the scope and spirit of the present disclosure. A freezer door 130 is arranged below refrigerator doors 128 for accessing freezer chamber 124. In the exemplary embodiment, freezer door 130 is coupled to a freezer drawer (not shown) slidably mounted within freezer chamber 124. An auxiliary door 127 may be coupled to an auxiliary drawer (not shown) which is slidably mounted within the auxiliary chamber (not shown).

Operation of the refrigerator appliance 100 can be regulated by a controller 134 that is operatively coupled to a user interface panel 136. User interface panel 136 provides selections for user manipulation of the operation of refrigerator appliance 100 to modify environmental conditions therein, such as temperature selections, etc.

In some embodiments, user interface panel 136 may be proximate a dispenser assembly 132. Panel 136 provides selections for user manipulation of the operation of refrigerator appliance 100 such as, e.g., temperature selections, selection of automatic or manual override humidity control (as described in more detail below), etc. In response to user manipulation of the user interface panel 136, the controller 134 operates various components of the refrigerator appliance 100. Operation of the refrigerator appliance 100 can be regulated by the controller 134, e.g., controller 134 may regulate operation of various components of the refrigerator appliance 100 in response to programming and/or user manipulation of the user interface panel 136.

The controller 134 may include a memory and one or more microprocessors, CPUs or the like, such as general or special purpose microprocessors operable to execute programming instructions or micro-control code associated with operation of refrigerator appliance 100. The memory may represent random access memory such as DRAM, or read only memory such as ROM or FLASH. In one embodiment, the processor executes programming instructions stored in memory. The memory may be a separate component from the processor or may be included onboard within the processor. It should be noted that controllers 134 as disclosed herein are capable of and may be operable to perform any methods and associated method steps as disclosed herein.

The controller 134 may be positioned in a variety of locations throughout refrigerator appliance 100. In the illustrated embodiment, the controller 134 may be located within the door 128. In such an embodiment, input/output (“I/O”) signals may be routed between the controller and various operational components of refrigerator appliance 100. In one embodiment, the user interface panel 136 may represent a general purpose I/O (“GPIO”) device or functional block. In one embodiment, the user interface 136 may include input components, such as one or more of a variety of electrical, mechanical or electro-mechanical input devices including rotary dials, push buttons, and touch pads. The user interface 136 may include a display component, such as a digital or analog display device designed to provide operational feedback to a user. For example, the user interface 136 may include a touchscreen providing both input and display functionality. The user interface 136 may be in communication with the controller via one or more signal lines or shared communication busses.

As may be seen in FIG. 3, a plurality of food storage elements, such as bins 138, shelves 142, and drawers 140 are disposed within the fresh food storage chamber 122. As will be described in more detail below, the drawers 140 may be configured for storing produce, such as fruits and vegetables, and, in particular, the refrigerator appliance may be operable

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to provide, and configured for providing, improved shelf life of the produce stored therein. Thus, the drawers **140** may also be referred to as produce drawers **140** or as a vegetable drawer **140** and a fruit drawer **140**. For example, in some embodiments, the refrigerator appliance may include two drawers **140**, e.g., as illustrated in FIG. 3.

Referring now generally to FIGS. 3 through 6, refrigerator appliance **100** may further include an inventory management system that is generally configured to monitor one or more chambers of refrigerator appliance **100** to monitor the status of inventory stored therein. More specifically, as described in more detail below, inventory management system may include one or more sniffers or sensors **200** (see, e.g., FIGS. 4 and 5), cameras **192** (see, e.g., FIGS. 3 and 5), or other detection devices that are used to monitor fresh food chamber **122**, and the drawers **140** in particular, to detect and monitor status of objects (e.g., identified generally by reference numeral **182** in FIG. 6) that are positioned in or removed from the drawers **140**. In this regard, inventory management system may use data from each of these devices to obtain a representation or knowledge of the identity, position, and/or other qualitative or quantitative characteristics of objects **182**, e.g., produce items such as fruits and/or vegetables, within the drawers **140**. Although the inventory management system is described herein as monitoring drawers **140** for the detection of objects **182**, it should be appreciated that aspects of the present subject matter may be used to monitor objects or items in any other suitable appliance, chamber, etc.

As shown schematically in FIG. 3, the inventory management system may include a camera assembly **190** that is generally positioned and configured for obtaining images of refrigerator appliance **100** during operation. Specifically, according to the illustrated embodiment, camera assembly **190** includes one or more cameras **192** that are mounted to cabinet **120**, to doors **128**, or are otherwise positioned in view of fresh food chamber **122**. As shown in FIG. 3, a camera **192** of camera assembly **190** is mounted to cabinet **120** at a front opening of fresh food chamber **122** and is oriented to have a field of view **194** directed across the front opening and/or into fresh food chamber **122**, and in particular into the drawers **140**, such as when the drawers **140** are in an open position, such as extended forward generally along the transverse direction T such that one or both of the drawers **140** extend beyond (forward of) a front edge of a shelf **142** or shelves **142** between the camera assembly **190** and the drawers **140**, such as above the drawers **140**.

Although a single camera **192** is illustrated in FIG. 3, it should be appreciated that camera assembly **190** may include a plurality of cameras **192** positioned within cabinet **120**, wherein each of the plurality of cameras **192** has a specified monitoring zone or range positioned around fresh food chamber **122**. In this regard, for example, the field of view **194** of each camera **192** may be limited to or focused on a specific area within fresh food chamber **122**, such as one camera **192** for each drawer **140**.

Notably, however, it may be desirable to position each camera **192** proximate the front opening of fresh food chamber **122** and orient each camera **192** such that the field of view **194** is directed into fresh food chamber **122**. In this manner, privacy concerns related to obtaining images of the user of the appliance **100** may be mitigated or avoided altogether. According to exemplary embodiments, camera assembly **190** may be used to facilitate an inventory management process for refrigerator appliance **100**. As such, each camera **192** may be positioned at an opening to fresh food chamber **122** to monitor food items (identified gener-

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ally as objects **182**) that are being added to or removed from fresh food chamber **122**, and in particular that are being placed in or removed from the drawers **140**.

It should be appreciated that according to alternative embodiments, camera assembly **190** may include any suitable number, type, size, and configuration of camera(s) **192** for obtaining images of any suitable areas or regions within or around refrigerator appliance **100**. In addition, it should be appreciated that each camera **192** may include features for adjusting the field of view and/or orientation.

It should be appreciated that the images obtained by camera assembly **190** may vary in number, frequency, angle, resolution, detail, etc. in order to improve the clarity of the particular regions surrounding or within refrigerator appliance **100**. In addition, according to exemplary embodiments, controller **134** may be configured for illuminating the chilled chamber using one or more light sources prior to obtaining images. Notably, controller **134** of refrigerator appliance **100** (or any other suitable dedicated controller) may be communicatively coupled to camera assembly **190** and may be programmed or configured for analyzing the images obtained by camera assembly **190**, e.g., in order to identify items being added or removed from refrigerator appliance **100**, as described in more detail below.

In general, controller **134** may be operably coupled to camera assembly **190** for analyzing one or more images obtained by camera assembly **190** to extract useful information regarding objects **182** located within the drawers **140**. In this regard, for example, images obtained by camera assembly **190** may be used to extract a barcode, identify a product, monitor the motion of the product, or obtain other product information related to object **182**. Notably, this analysis may be performed locally (e.g., on controller **134**) or may be transmitted to a remote server (e.g., in the “cloud,” as those of ordinary skill in the art will recognize as referring to a remote server or database in a distributed computing environment including at least one remote server and the local controller **134**) for analysis. Such analysis is intended to facilitate inventory management, e.g., by identifying a food item being added to or removed from the fresh food chamber **122**.

Specifically, according to an exemplary embodiment, camera **192** (or multiple cameras **192** in the camera assembly **190** collectively) is oriented down from a top center of cabinet **120** and has a field of view **194** (e.g., as shown schematically in FIG. 3 and corresponding to the exemplary image of FIG. 6) that covers a width of fresh food chamber **122**, e.g., a collective width of both drawers **140**. In this manner, the field of view **194** of camera **192**, and the resulting images obtained, may capture any motion or movement of an object into and/or out of drawers **140**. The images obtained by camera assembly **190** may include one or more still images, one or more video clips, or any other suitable type and number of images suitable for identification of food items (e.g., identified generally by reference numeral **182**) or inventory analysis.

Notably, camera assembly **190** may obtain images upon any suitable trigger, such as a time-based imaging schedule where camera assembly **190** periodically images and monitors the drawers **140**. According to still other embodiments, camera assembly **190** may periodically take low-resolution images until motion (such as opening, e.g., sliding forward, of one or both drawers **140**) is detected (e.g., via image differentiation of low-resolution images), at which time one or more high-resolution images may be obtained. According to still other embodiments, refrigerator appliance **100** may include one or more motion sensors (e.g., optical, acoustic,

electromagnetic, etc.) that are triggered when an object **182** is being added to or removed from the drawers **140**, and camera assembly **190** may be operably coupled to such motion sensors to obtain images of the object **182** during such movement.

According to still other embodiments, refrigerator appliance **100** may include a door switch that detects when refrigerator door **128** is opened, at which point camera assembly **190** may begin obtaining one or more images. According to exemplary embodiments, the image **300** may be obtained continuously or periodically while refrigerator doors **128** are open and/or while one or both drawers **140** are in the open position. In this regard, obtaining image **300** may include determining that a door and/or drawer of the refrigerator appliance is open and capturing images at a set frame rate while the door and/or drawer is open.

Notably, the motion of the food items between image frames may be used to determine whether the food item **182** is being removed from or added into fresh food chamber **122**. It should be appreciated that the images obtained by camera assembly **190** may vary in number, frequency, angle, resolution, detail, etc. in order to improve the clarity of food items **182**. In addition, according to exemplary embodiments, controller **134** may be configured for illuminating a refrigerator light (not shown) while obtaining image **300**. Other suitable imaging triggers are possible and within the scope of the present subject matter.

As illustrated in FIGS. 4 and 5, in various embodiments, the refrigerator appliance **100** may include an atmospheric condition sensor or sniffer **200** in fluid communication with the fresh food chamber **122**. For example, the sensor **200** may be positioned and configured within the housing **120**, such as within the fresh food chamber **122** defined therein, such as within the drawer **140** in the fresh food chamber **122**, such that fluid, e.g., gases, such as air or other atmospheric gases within the fresh food chamber **122** (and in particular within the drawer **140**, such as within the food storage compartment **144** defined therein) flow to and around and/or across the sensor **200**, whereby the sensor **200** may detect or monitor atmospheric conditions, such as atmospheric composition, temperature, humidity, and other similar atmospheric conditions within the fresh food chamber **122** and the drawer **140**.

In some embodiments, multiple sensors **200** may be provided. For example, when sensor **200** is positioned in one of the drawers **140**, another sensor **200** may be provided in the other drawer **140**. As another example, multiple sensors **200** may be provided, each operable to and configured to measure different atmospheric conditions, such as a temperature sensor and a chemical sensor, e.g., the chemical sensor may be a sniffer that detects or measures a concentration of a particular chemical or type of chemicals, such as ethylene.

Referring now to FIGS. 7 and 8, an exemplary drawer **140** may include a food storage compartment **144**. The food storage compartment **144** may be defined by a plurality of walls of the drawer **140**. For example, the plurality of walls may include a front wall **146**, a back wall **148**, a left wall **150**, and a right wall **152**. Directional terms such as “left” and “right” are used herein with reference to the perspective of a user standing in front of the refrigerator appliance **100** to access items stored therein. One of the walls, e.g., the front wall **146** as in the illustrated example embodiment of FIG. 7, may include one or more vent apertures **154** defined in and through the wall, e.g., the front wall **146**. In such embodiments, the drawer **140** may include a humidity control knob, slider, or lever which adjusts the degree of

openness of the vent aperture **154**, such as a slider **156**, e.g., as illustrated in FIG. 8. The slider **156** may be movable to selectively vary a degree of obstruction of the one or more vent apertures **154**, such as between an open position where the one or more vent apertures are unobstructed to provide increased air circulation into and through the food storage compartment **144** and a closed position where the one or more vent apertures are obstructed or blocked by the slider **156** to restrict air flow into the food storage compartment **144**.

For example, as illustrated in FIG. 8, the slider **156** may be movable in two generally opposing directions along a single line, e.g., along direction **1000** as indicated in FIG. 8, within a track **158**. Additionally, those of ordinary skill in the art will recognize that the slider **156** may be movable through a plurality of intermediate positions between the open position and the closed position, such as the exemplary intermediate position illustrated in FIG. 8, where the one or more vent apertures are partially obstructed by the slider **156**. Thus, the slider **156** may be movable within the track **158** in two opposing directions, e.g., back and forth, along direction **1000** between the open and closed positions and through a plurality of intermediate positions therebetween. In particular, the exemplary embodiment illustrated in FIG. 8 includes multiple vent apertures **154** and the slider **156** in an exemplary intermediate position where one vent aperture **154** is completely unobstructed or fully open and another adjacent vent aperture **154** is partially open, e.g., is partially obstructed.

Varying the position of the slider **156** as described provides varying levels of humidity within the food storage compartment **144** of the drawer **140**. For example, when the slider is in the closed position or in an intermediate position, humidity (e.g., air having a relatively high moisture content as compared to ambient air outside of the refrigerator appliance and/or air within the remainder of the fresh food chamber **122** outside of the drawer **140**) may accumulate within the food storage compartment **144** of the drawer **140**, whereas when the slider **156** is in the open position, the humidity level within the drawer **140** will equilibrate with the ambient humidity level, e.g., within the remainder of the fresh food chamber **122** outside of the drawer **140**. Further, the various intermediate positions provide a varying rate at which the humidity within the food storage compartment **144** of the drawer **140** will reach equilibrium with the humidity in the remainder of the fresh food chamber **122** outside of the drawer **140**, such as faster equilibration (and resultingly lower humidity within the food storage compartment **144** of the drawer **140**) as the intermediate position of the slider **156** approaches the open position and slower equilibration (and resultingly higher humidity within the food storage compartment **144** of the drawer **140**) as the intermediate position of the slider **156** approaches the closed position. For example, the intermediate position illustrated in FIG. 8, with one vent aperture **154** open and another partially open and partially closed, permits some moisture to escape from the atmosphere within the food storage compartment **144** while also retaining a portion of the humidity, e.g., for produce items which prefer a moderate humidity level for storage.

Using the teachings disclosed herein, one of skill in the art will understand that the present subject matter can be used with other types of refrigerators such as a refrigerator/freezer combination, side-by-side, bottom mount, compact, and any other style or model of refrigerator appliance. Accordingly, other configurations of refrigerator appliance **100** could be provided, it being understood that the configu-

rations shown in the accompanying FIGS. and the description set forth herein are by way of example for illustrative purposes only.

Now that the construction and configuration of refrigerator appliance **100** and camera assembly **190** have been presented according to an exemplary embodiment of the present subject matter, exemplary methods for operating a refrigerator appliance, such as refrigerator appliance **100**, are provided. Such methods may also be used to operate a camera assembly, e.g., camera assembly **190**, or any other suitable camera assembly for monitoring appliance operation or inventory. In this regard, for example, controller **134** may be configured for implementing one or more of the following exemplary methods. However, it should be appreciated that the exemplary methods are discussed herein only to describe exemplary aspects of the present subject matter, and are not intended to be limiting.

Turning now to FIG. **9**, embodiments of the present disclosure may include a method **400** of operating a refrigerator appliance, such as the exemplary refrigerator appliance **100** described above. For example, the refrigerator appliance may include a controller and a cabinet defining a food storage chamber with a drawer slidably mounted within the food storage chamber. The drawer may be slidable between a closed position and an open position. The drawer may include a plurality of walls defining a food storage compartment. Also by way of example, the refrigerator may further include a sensor operable to detect an atmospheric condition within the food storage chamber of the drawer and a camera assembly positioned and configured for monitoring the drawer, as described above.

As shown in FIG. **9**, method **400** includes, at step **410**, obtaining an image of a chilled chamber of a refrigerator appliance using a camera assembly. For example, camera assembly **190** of refrigerator appliance **100** may obtain an image **300** (e.g., as shown for example in FIG. **6**) within fresh food chamber **122** and/or food storage chamber **144** of drawer **140**, which may include in its field of view a plurality of objects **182**. In this regard, camera assembly **190** of refrigerator appliance **100** may obtain one or more images (e.g., such as image **300**) of fresh food chamber **122**, freezer chamber **124**, or any other zone or region within or around refrigerator appliance **100**.

In some embodiments, the method may also include, and/or the refrigerator appliance may further be configured for, identifying one or more food items, such as identifying a first food item and a second food item, e.g., based on one or more images. In some embodiments, identification of the food item may be accomplished with the camera assembly **190**. For example, the refrigerator appliance may include a camera, and the step of identifying the food item may include identifying the food item based on an image captured by the camera. In some embodiments, the operation of the camera may be tied to the door opening, e.g., the camera may be operable and configured to capture an image each time the door is opened and/or each time the door is closed after detecting a door opening. The structure and operation of cameras are understood by those of ordinary skill in the art and, as such, the camera is not illustrated or described in further detail herein for the sake of brevity and clarity. In such embodiments, the controller **134** of the refrigerator appliance **100** may be configured for image-based processing, e.g., to identify a food item based on an image of the food item, e.g., a photograph of the food item taken with the camera(s) **192** of the camera assembly **190**. For example, the controller **134** may be configured to identify the food item

by comparison of the image to a stored image of a known or previously-identified food item.

In the exemplary embodiment illustrated in FIG. **9**, the method **400** may include a step **420** of analyzing the image that was obtained at step **410** to identify a first food item and a second food item in the food storage chamber of the drawer. For example, controller **134** of refrigerator appliance **100** (or any other suitable dedicated controller) may be communicatively coupled to camera assembly **190** and may be programmed or configured for analyzing the images obtained by camera assembly **190**, e.g., in order to identify items stored in refrigerator appliance **100**, as described in more detail above.

Step **420** includes analyzing the image using to identify objects, e.g., at least a first food item and a second food item, which are disposed in the food storage chamber **144** of the drawer **140**. It should be appreciated that this analysis may utilize any suitable image analysis techniques, image decomposition, image segmentation, image processing, etc. This analysis may be performed entirely by controller **134**, may be offloaded to a remote server for analysis, may be analyzed with user assistance (e.g., via user interface panel **136**), or may be analyzed in any other suitable manner. According to exemplary embodiments of the present subject matter, the analysis performed at step **420** may include a machine learning image recognition process.

According to exemplary embodiments, this image analysis may use any suitable image processing technique, image recognition process, etc. As used herein, the terms “image analysis” and the like may be used generally to refer to any suitable method of observation, analysis, image decomposition, feature extraction, image classification, etc. of one or more images, videos, or other visual representations of an object. As explained in more detail below, this image analysis may include the implementation of image processing techniques, image recognition techniques, or any suitable combination thereof. In this regard, the image analysis may use any suitable image analysis software or algorithm to constantly or periodically monitor objects within fresh food chamber **122**, such as within a drawer **140** therein. It should be appreciated that this image analysis or processing may be performed locally (e.g., by controller **134**) or remotely (e.g., by offloading image data to a remote server or network, e.g., in the cloud).

Specifically, the analysis of the one or more images may include implementation an image processing algorithm. As used herein, the terms “image processing” and the like are generally intended to refer to any suitable methods or algorithms for analyzing images that do not rely on artificial intelligence or machine learning techniques (e.g., in contrast to the machine learning image recognition processes described below). For example, the image processing algorithm may rely on image differentiation, e.g., such as a pixel-by-pixel comparison of two sequential images. This comparison may help identify substantial differences between the sequentially obtained images, e.g., to identify movement, the presence of a particular object, the existence of a certain condition, etc. For example, one or more reference images may be obtained when a particular condition exists, and these reference images may be stored for future comparison with images obtained during appliance operation. Similarities and/or differences between the reference image and the obtained image may be used to extract useful information for improving appliance performance. For example, image differentiation may be used to determine when a pixel level motion metric passes a predetermined motion threshold.

The processing algorithm may further include measures for isolating or eliminating noise in the image comparison, e.g., due to image resolution, data transmission errors, inconsistent lighting, or other imaging errors. By eliminating such noise, the image processing algorithms may improve accurate object detection, avoid erroneous object detection, and isolate the important object, region, or pattern within an image. In addition, or alternatively, the image processing algorithms may use other suitable techniques for recognizing or identifying particular items or objects, such as edge matching, divide-and-conquer searching, greyscale matching, histograms of receptive field responses, or another suitable routine (e.g., executed at the controller 134 based on one or more captured images from one or more cameras). Other image processing techniques are possible and within the scope of the present subject matter.

In addition to the image processing techniques described above, the image analysis may include utilizing artificial intelligence (“AI”), such as a machine learning image recognition process, a neural network classification module, any other suitable artificial intelligence (AI) technique, and/or any other suitable image analysis techniques, examples of which will be described in more detail below. Moreover, each of the exemplary image analysis or evaluation processes described below may be used independently, collectively, or interchangeably to extract detailed information regarding the images being analyzed to facilitate performance of one or more methods described herein or to otherwise improve appliance operation. According to exemplary embodiments, any suitable number and combination of image processing, image recognition, or other image analysis techniques may be used to obtain an accurate analysis of the obtained images.

In this regard, the image recognition process may use any suitable artificial intelligence technique, for example, any suitable machine learning technique, or for example, any suitable deep learning technique. According to an exemplary embodiment, the image recognition process may include the implementation of a form of image recognition called region based convolutional neural network (“R-CNN”) image recognition. Generally speaking, R-CNN may include taking an input image and extracting region proposals that include a potential object or region of an image. In this regard, a “region proposal” may be one or more regions in an image that could belong to a particular object or may include adjacent regions that share common pixel characteristics. A convolutional neural network is then used to compute features from the region proposals and the extracted features will then be used to determine a classification for each particular region.

According to still other embodiments, an image segmentation process may be used along with the R-CNN image recognition. In general, image segmentation creates a pixel-based mask for each object in an image and provides a more detailed or granular understanding of the various objects within a given image. In this regard, instead of processing an entire image—i.e., a large collection of pixels, many of which might not contain useful information—image segmentation may involve dividing an image into segments (e.g., into groups of pixels containing similar attributes) that may be analyzed independently or in parallel to obtain a more detailed representation of the object or objects in an image. This may be referred to herein as “mask R-CNN” and the like, as opposed to a regular R-CNN architecture. For example, mask R-CNN may be based on fast R-CNN which is slightly different than R-CNN. For example, R-CNN first applies a convolutional neural network (“CNN”) and then

allocates it to zone recommendations on the covn5 property map instead of the initially split into zone recommendations. In addition, according to exemplary embodiments, standard CNN may be used to obtain, identify, or detect any other qualitative or quantitative data related to one or more objects or regions within the one or more images. In addition, a K-means algorithm may be used.

According to still other embodiments, the image recognition process may use any other suitable neural network process while remaining within the scope of the present subject matter. For example, the step of analyzing the one or more images may include using a deep belief network (“DBN”) image recognition process. A DBN image recognition process may generally include stacking many individual unsupervised networks that use each network’s hidden layer as the input for the next layer. According to still other embodiments, the step of analyzing one or more images may include the implementation of a deep neural network (“DNN”) image recognition process, which generally includes the use of a neural network (computing systems inspired by the biological neural networks) with multiple layers between input and output. Other suitable image recognition processes, neural network processes, artificial intelligence analysis techniques, and combinations of the above described or other known methods may be used while remaining within the scope of the present subject matter.

In addition, it should be appreciated that various transfer techniques may be used but use of such techniques is not required. If using transfer techniques learning, a neural network architecture may be pretrained such as VGG16/VGG19/ResNet50 with a public dataset then the last layer may be retrained with an appliance specific dataset. In addition, or alternatively, the image recognition process may include detection of certain conditions based on comparison of initial conditions, may rely on image subtraction techniques, image stacking techniques, image concatenation, etc. For example, the subtracted image may be used to train a neural network with multiple classes for future comparison and image classification.

It should be appreciated that the machine learning image recognition models may be actively trained by the appliance with new images, may be supplied with training data from the manufacturer or from another remote source, or may be trained in any other suitable manner. For example, according to exemplary embodiments, this image recognition process relies at least in part on a neural network trained with a plurality of images of the appliance in different configurations, experiencing different conditions, or being interacted with in different manners. This training data may be stored locally or remotely and may be communicated to a remote server for training other appliances and models.

It should be appreciated that image processing and machine learning image recognition processes may be used together to facilitate improved image analysis, object detection, or to extract other useful qualitative or quantitative data or information from the one or more images that may be used to improve the operation or performance of the appliance. Indeed, the methods described herein may use any or all of these techniques interchangeably to improve image analysis process and facilitate improved appliance performance and consumer satisfaction. The image processing algorithms and machine learning image recognition processes described herein are only exemplary and are not intended to limit the scope of the present subject matter in any manner.

Referring again to FIG. 9 in particular, the exemplary method 400 may further include a step 430 of detecting that

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an atmospheric condition within the food storage chamber, e.g., food storage chamber **144**, above a predetermined threshold. The predetermined threshold may, for example, be a default value stored in a memory of the controller. Atmospheric conditions may be monitored and/or detected with one or more sniffers or sensors **200**, as described above. The predetermined threshold may be, for example, an ethylene level. Additional exemplary atmospheric conditions, and corresponding predetermined thresholds therefor, include a temperature, a humidity level, and/or a level or concentration of any other chemical or component in the atmosphere within the drawer, e.g., drawer **140**.

Method **400** may also include a step **440** of identifying, based on the analysis of the image, one of the first food item and the second food item as the source of the atmospheric condition above the predetermined threshold. For example, the steps **420** and **440** may use the same image or multiple images from the same set of images, where the set of images includes multiple images of the same area or location taken over time. For example, the identification of one of the first food item and the second food item as the source of the atmospheric condition may include image analysis whereby a change in color in a food item, such as darkening or turning brown, etc., of a fruit item, vegetable item, or other similar produce item is recognized from a chronological series of images of the same objects in the drawer.

Also, it should be understood that the steps are not necessarily performed in a given order, e.g., the detection step **430** may occur prior to the identification of the first and second food items, such as the identification of the food items may occur in response to detecting an atmospheric condition above the predetermined threshold. As but one example of many possible examples, the atmospheric condition may be an ethylene level, and the predetermined threshold may be an excessive ethylene level, e.g., which may be excessive in that the level of ethylene may be detrimental to the storage of at least one produce item, where the method may thus include detecting an excessive ethylene level and, in response to detecting the ethylene level, obtaining and analyzing an image to locate the source of the ethylene level.

In some embodiments, method **400** may further include providing a user notification. The user notification may include an indication or identification of the one of the first food item and the second food item that was identified as the source of the atmospheric condition above the predetermined threshold.

In some embodiments, the sensor may be operable to detect the atmospheric condition within the food storage chamber of the drawer when the drawer is in the closed position, such as step **430** may be performed when the drawer is in the closed position. For example, the controller may ping or query the sensor when the drawer is in the closed position, where the closed position may be detected by the controller based on a position switch or position sensor (such as a Hall effect sensor) and/or based on an image from the camera assembly, where the controller may analyze such image to recognize and detect when the drawer is in the closed position. Measuring or detecting the atmospheric condition while the drawer is in the closed position may advantageously provide a more accurate reading of the atmosphere within the drawer itself, such as in the food storage compartment therein, as opposed to ambient conditions outside of the drawer, e.g., in the remainder of the fresh food compartment and/or outside of the refrigerator appliance.

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In some embodiments, the camera assembly may be positioned and configured for monitoring the food storage chamber of the drawer when the drawer is in the open position. For example, images of the contents within the drawer may be advantageously clearer, e.g., less obstructed, when the images are obtained while the drawer is in the open position, such as the drawer may be extended outward away from other contents and structures, e.g., shelves, in the refrigerator when the drawer is in the open position, thereby allowing a clearer and more complete observation of the interior of the drawer and contents thereof when the drawer is in the open position.

In some embodiments, the drawer may also include a vent aperture defined in and through one of the plurality of walls. In such embodiments, the camera assembly may be positioned and configured for monitoring the vent aperture of the drawer, e.g., the vent aperture may be positioned within the field of view of the camera. Such embodiments may also include determining an optimal humidity level for at least one of the first food item and the second food item and determining an optimal position for a slider at the vent aperture corresponding to the determined optimal humidity. Exemplary embodiments in which the camera assembly is positioned and configured for monitoring the vent aperture of the drawer may further include analyzing the image to determine whether the slider at the vent aperture is in the optimal position, and providing a user notification when the slider at the vent aperture is not in the optimal position, where the user notification may be, e.g., audible and/or visual, as described in more detail below, and may be provided locally and/or remotely, also as described in more detail below.

Turning now to FIG. **10**, embodiments of the present disclosure may include a method **500** of operating a refrigerator appliance, such as the exemplary refrigerator appliance **100** described above. For example, the refrigerator appliance may include a controller and a plurality of food storage drawers, etc., as described above.

Method **500** also includes an image obtaining step **510** and an analysis and identification step **520**, which are similar to steps **410** and **420** described above, and such description is not repeated for the sake of brevity.

In contrast to the predetermined threshold described above with respect to exemplary method **400**, method **500** may include defining new or additional thresholds for one or more atmospheric conditions in the refrigerator appliance, such as in the drawer **140**. For example, the one or more thresholds may be based on and/or in response to the identified food items, such as an expected or anticipated ethylene level for a particular type of produce item, where the expected or anticipated ethylene level corresponds to the identified food item being ripe (or overripe, etc.). Thus, in some embodiments, exemplary method **500** may include a step **530** of setting a first threshold for an atmospheric condition based on the identity of the first food item and a step **540** of setting a second threshold for the atmospheric condition based on the identity of the second food item. In at least some embodiments, the first food item may be different from the second food item, and thus the first threshold may also be different from the second threshold, although different food items may not necessarily have different thresholds.

As illustrated in FIG. **10**, method **500** may further include a step **550** of monitoring the atmospheric condition for which the first and second thresholds were set at steps **530** and **540**. Such monitoring may be performed at least in part by the sensor, such as by the controller of the refrigerator

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appliance which is in operative communication with the sensor, e.g., exemplary methods may include monitoring the atmospheric condition using the sensor and/or by the sensor.

Still referring to FIG. 10, method 500 may also include steps of providing one or more user notifications. Such notifications may be provided locally, e.g., on the user interface panel 136 of the refrigerator appliance 100, and/or remotely, such as on a remote device not directly physically attached or connected to the refrigerator appliance, e.g., a smartphone, smart home system, or other similar device. The user notification may include one or more of a visual notification, e.g., illuminating an indicator light or providing a text notification, and/or an audible notification, such as a chime or alert tone, etc. For example, method 500 may include a step 560 of providing a first user notification when the first atmospheric condition threshold based on the identity of the first food item is reached, and a step 570 of providing a second user notification when the second atmospheric condition threshold based on the identity of the second food item is reached. Thus, for example, customized and responsive monitoring and inventory management may be provided in method 500, where each food item is tracked separately and specifically based on atmospheric conditions which are more significant or sensitive for the particular identified food item.

In some embodiments, the first atmospheric condition threshold of step 530 may include a first ethylene level and the second atmospheric condition threshold of step 540 may include a second ethylene level.

In some embodiments, the sensor may be operable to detect the atmospheric condition within the food storage chamber of the drawer when the drawer is in the closed position. For example, the controller may ping or query the sensor when the drawer is in the closed position, where the closed position may be detected by the controller based on a position switch or position sensor (such as a Hall effect sensor) and/or based on an image from the camera assembly, where the controller may analyze such image to recognize and detect when the drawer is in the closed position. Measuring or detecting the atmospheric condition while the drawer is in the closed position may advantageously provide a more accurate reading of the atmosphere within the drawer itself, such as in the food storage compartment therein, as opposed to ambient conditions outside of the drawer, e.g., in the remainder of the fresh food compartment and/or outside of the refrigerator appliance.

In some embodiments, the camera assembly may be positioned and configured for monitoring the food storage chamber of the drawer when the drawer is in the open position. For example, images of the contents within the drawer may be advantageously clearer, e.g., less obstructed, when the images are obtained while the drawer is in the open position, such as the drawer may be extended outward away from other contents and structures, e.g., shelves, in the refrigerator when the drawer is in the open position, thereby allowing a clearer and more complete observation of the interior of the drawer when the drawer is in the open position.

In some embodiments, the drawer may also include a vent aperture defined in and through one of the plurality of walls. In such embodiments, the camera assembly may be positioned and configured for monitoring the vent aperture of the drawer, e.g., the vent aperture may be positioned within the field of view of the camera. Such embodiments may also include determining an optimal humidity level for at least one of the first food item and the second food item and determining an optimal position for a slider at the vent

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aperture corresponding to the determined optimal humidity. Further, such embodiments may also or instead include a humidity level as the first and second thresholds for the atmospheric condition, e.g., the atmospheric condition may be humidity, and the first and second thresholds may each be a humidity level. Exemplary embodiments in which the camera assembly is positioned and configured for monitoring the vent aperture of the drawer may further include analyzing the image to determine whether the slider at the vent aperture is in the optimal position, and providing a user notification when the slider at the vent aperture is not in the optimal position, where the user notification may be, e.g., audible and/or visual, as described above, and may be provided locally and/or remotely, also as described above.

Turning now to FIG. 11, embodiments of the present disclosure may also include a method 600 of operating a refrigerator appliance, such as the exemplary refrigerator appliance 100 described above. For example, the refrigerator appliance may include a controller and a plurality of food storage drawers, etc., as described above.

Method 600 also includes an image obtaining step 610 and an analysis and identification step 620, which are similar to steps 410/510 and 420/520 described above, and such description is not repeated for the sake of brevity.

Method 600 may also include a step 630 of determining that the first food item and the second food item are incompatible for co-storage. For example, such incompatibility may include different optimal humidity levels and/or temperatures. As another example, such incompatibility may also or instead include one of the first food item and the second food item being an ethylene producer, e.g., which generates or gives off a high amount of ethylene (as compared to other produce items) as it ripens or ages, and the other of the first food item and the second food item being ethylene sensitive, e.g., where exposure to levels of ethylene given off by the one food item may accelerate the aging rate of the other food item. In such embodiments, the determination of incompatibility may be based at least in part on an ethylene production rate of one of the first food item and the second food item.

After determining that the first food item and the second food item are incompatible for co-storage, and in response to such determination, the method 600 may then include a step 640 of providing a user notification including a recommendation to relocate one of the first food item and the second food item. For example, when more than one drawer is included in the refrigerator appliance, the recommendation may include recommending to move the one food item to another drawer. As another example, the recommendation may include recommending to move the one food item to another part of the fresh food chamber, e.g., outside of the drawer or drawers, or may include recommending to store the one food item at room temperature, e.g., outside of the refrigerator appliance. Further, such examples are not required, e.g., the user notification may simply provide a recommendation to remove or relocate the one food item from the drawer without specifying to where the one food item should be moved.

This written description uses examples to disclose the invention, including the best mode, and also to enable any person skilled in the art to practice the invention, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the invention is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they include structural elements that do not differ from the

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literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal languages of the claims.

What is claimed is:

1. A method of operating a refrigerator appliance, the refrigerator appliance comprising a cabinet defining a food storage chamber with a drawer slidably mounted within the food storage chamber whereby the drawer is slidable between a closed position and an open position, the drawer comprising a plurality of walls defining a food storage compartment and a vent aperture defined in and through one of the plurality of walls, a sensor operable to detect an atmospheric condition within the food storage chamber of the drawer, and a camera assembly positioned and configured for monitoring the vent aperture of the drawer, the method comprising:

obtaining an image using the camera assembly;
analyzing the image to identify a first food item and a second food item in the food storage chamber of the drawer;
detecting, with the sensor, an atmospheric condition within the food storage chamber above a predetermined threshold;
identifying, based on the analysis of the image, one of the first food item and the second food item as the source of the atmospheric condition above the predetermined threshold;
determining an optimal humidity level for at least one of the first food item and the second food item;
determining an optimal position for a slider at the vent aperture corresponding to the determined optimal humidity;
analyzing the image to determine whether the slider at the vent aperture is in the optimal position; and
providing a user notification when the slider at the vent aperture is not in the optimal position.

2. The method of claim 1, wherein the atmospheric condition comprises an ethylene level.

3. The method of claim 1, wherein the sensor is operable to detect the atmospheric condition within the food storage chamber of the drawer when the drawer is in the closed position.

4. The method of claim 1, wherein the camera assembly is positioned and configured for monitoring the food storage chamber of the drawer when the drawer is in the open position.

5. The method of claim 1, wherein the atmospheric condition detected with the sensor is a first atmospheric condition, further comprising setting a first threshold for a second atmospheric condition based on the identity of the first food item, setting a second threshold for the second atmospheric condition based on the identity of the second food item, monitoring the second atmospheric condition by the sensor, providing a first user notification when the second atmospheric condition reaches the first threshold, and providing a second user notification when the second atmospheric condition reaches the second threshold.

6. A method of operating a refrigerator appliance, the refrigerator appliance comprising a cabinet defining a food storage chamber with a drawer slidably mounted within the food storage chamber whereby the drawer is slidable between a closed position and an open position, the drawer comprising a plurality of walls defining a food storage compartment and a vent aperture defined in and through one of the plurality of walls, a sensor operable to detect an atmospheric condition within the food storage chamber of

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the drawer, and a camera assembly positioned and configured for monitoring the vent aperture of the drawer, the method comprising:

obtaining an image using the camera assembly;
analyzing the image to identify a first food item and a second food item in the food storage chamber of the drawer;
setting a first threshold for an atmospheric condition based on the identity of the first food item;
setting a second threshold for the atmospheric condition based on the identity of the second food item;
monitoring the atmospheric condition by the sensor;
providing a first user notification when the atmospheric condition reaches the first threshold;
providing a second user notification when the atmospheric condition reaches the second threshold;
determining an optimal humidity level for at least one of the first food item and the second food item;
determining an optimal position for a slider at the vent aperture corresponding to the determined optimal humidity;
analyzing the image to determine whether the slider at the vent aperture is in the optimal position; and
providing a third user notification when the slider at the vent aperture is not in the optimal position.

7. The method of claim 6, wherein the first threshold comprises a first ethylene level and the second threshold comprises a second ethylene level.

8. The method of claim 6, wherein the sensor is operable to detect the atmospheric condition within the food storage chamber of the drawer when the drawer is in the closed position.

9. The method of claim 6, wherein the camera assembly is positioned and configured for monitoring the food storage chamber of the drawer when the drawer is in the open position.

10. A method of operating a refrigerator appliance, the refrigerator appliance comprising a cabinet defining a food storage chamber with a drawer slidably mounted within the food storage chamber whereby the drawer is slidable between a closed position and an open position, the drawer comprising a plurality of walls defining a food storage compartment and a vent aperture defined in and through one of the plurality of walls, a sensor operable to detect an atmospheric condition within the food storage chamber of the drawer, and a camera assembly positioned and configured for monitoring the vent aperture of the drawer, the method comprising:

obtaining an image using the camera assembly;
analyzing the image to identify a first food item and a second food item in the food storage chamber of the drawer;
determining that the first food item and the second food item are incompatible for co-storage;
providing a user notification including a recommendation to relocate one of the first food item and the second food item;
determining an optimal humidity level for the other of the first food item and the second food item;
determining an optimal position for a slider at the vent aperture corresponding to the determined optimal humidity;
analyzing the image to determine whether the slider at the vent aperture is in the optimal position; and
providing a user notification when the slider at the vent aperture is not in the optimal position.

11. The method of claim 10, wherein the determination of incompatibility is based at least in part on an ethylene production rate of one of the first food item and the second food item.

12. The method of claim 10, wherein the sensor is operable to detect the atmospheric condition within the food storage chamber of the drawer when the drawer is in the closed position.

13. The method of claim 10, wherein the camera assembly is positioned and configured for monitoring the food storage chamber of the drawer when the drawer is in the open position.

14. The method of claim 10, wherein analyzing the image further comprises identifying a third food item in the food storage chamber of the drawer, further comprising setting a first threshold for an atmospheric condition based on the identity of the other of the first food item and the second food item, setting a second threshold for the atmospheric condition based on the identity of the third food item, monitoring the atmospheric condition by the sensor, providing a first user notification when the atmospheric condition reaches the first threshold, and providing a second user notification when the atmospheric condition reaches the second threshold.

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